

Python Programs – Tuple, Set and Dictionary

1. Tuple – PASS Results

Question

A trainer stores result statuses in a tuple. Write a Python program to find how many `PASS` results exist and find the first position of `PASS` using tuple methods.

Program

```
results = ("PASS", "FAIL", "PASS", "FAIL", "PASS", "PASS")

# Using tuple methods for both operations
pass_count = results.count("PASS")
first_position = results.index("PASS")

print("Number of PASS results:", pass_count)
print("First position of PASS:", first_position)
```

Sample Output

```
Number of PASS results: 4
First position of PASS: 0
```

Methods Used

- `count()` – Counts the number of occurrences of "PASS".
 - `index()` – Finds the first position of "PASS".
-

2. Set – Student Skills

Question

A student already has some skills. Write a Python program to add a new skill and several new skills using `add()` and `update()`. Also demonstrate `union()`, `intersection()`, and `difference()` operations using sets.

Program

```
# Set of skills already known by the student
skills = {"Python", "Java", "HTML"}

# Add one new skill using add()
skills.add("CSS")
```

```
# Add several new skills using update()
skills.update({"SQL", "C++", "JavaScript"})

print("Student Skills:", skills)

# Another set of skills
new_skills = {"Python", "SQL", "React", "JavaScript"}

# Union
print("Union:", skills.union(new_skills))

# Intersection
print("Intersection:", skills.intersection(new_skills))

# Difference
print("Difference:", skills.difference(new_skills))
```

Sample Output

Note: Sets are unordered in Python, so the order of items may vary.

```
Student Skills: {'Python', 'Java', 'HTML', 'CSS', 'SQL', 'C++',
'JavaScript'}

Union: {'Python', 'Java', 'HTML', 'CSS', 'SQL', 'C++', 'JavaScript',
'React'}

Intersection: {'Python', 'SQL', 'JavaScript'}

Difference: {'Java', 'HTML', 'CSS', 'C++'}
```

Methods Used

- `add()` – Adds one item to a set.
- `update()` – Adds multiple items to a set.
- `union()` – Combines the elements of two sets.
- `intersection()` – Finds common elements in two sets.
- `difference()` – Finds elements present in the first set but not in the second set.

3. Dictionary – Product Details

Question

An online store stores product codes and product names in a dictionary. Write a Python program to display all product codes using `keys()` and all product names using `values()`.

Program

```
products = {
    101: "Laptop",
```

```

        102: "Mobile Phone",
        103: "Headphones",
        104: "Keyboard"
    }

    # Display all product codes using keys()
    print("Product Codes:")
    for code in products.keys():
        print(code)

    # Display all product names using values()
    print("\nProduct Names:")
    for name in products.values():
        print(name)

```

Sample Output

```

Product Codes:
101
102
103
104

```

```

Product Names:
Laptop
Mobile Phone
Headphones
Keyboard

```

Methods Used

- `keys()` – Returns all the keys in the dictionary.
- `values()` – Returns all the values in the dictionary.

Summary

Data Type

Methods Used

Tuple `count()`, `index()`

Set `add()`, `update()`, `union()`, `intersection()`, `difference()`

Dictionary `keys()`, `values()`

These programs demonstrate basic Python operations on **Tuple**, **Set**, and **Dictionary** data types.